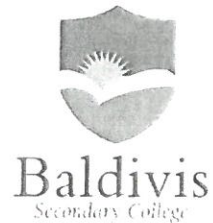


YEAR 8 Algebra
Semester 1 2017
Mini Test 2



Name: SOLUTIONS

Total Marks: 55 marks
Total Time: 55 minutes

**Full working out must be shown to get full marks.
Attempt all questions**

**Resources allowed:
1 A4 page, (1 side) of hand written notes
Calculator**

1. For the expression $6xy + 2x - 4y^2 + 3$, state: (4)
 - a) the coefficient of $x = 2$
 - b) the constant term $= 3$
 - c) the number of terms $= 4$
 - d) the coefficient of $xy = 6$

2. State whether each of the following is true or false: (5)
 - a) The constant term in the expression $5x + 7$ is 5. *False*
 - b) $16xy$ and $5yx$ are like terms. *True*
 - c) The coefficient of d in the expression $6d^2 + 7d + 8abd + 3$ is 7. *True*
 - d) The highest common factor of $12abc$ and $16c$ is $2c$. *false*
 - e) The coefficient of x and y in the expression $2x + 2y$ is 2. *True*

3. Circle the **like terms** in each group (5)
 - a) $2p, 3k, p$
 - b) $9j, 3z, -2z$
 - c) $100d, 3d, 9e$
 - d) $14r^2, 14r, 2r^2$
 - e) $-6w, 4q, -2w$

(4)

4. Simplify the following. .

a) $4a + 3a = 7a$

b) $9i - 4i = 5i$

c) $12d - 15d = -3d$

d) $16g - 9g + 4g = 11g$

(4)

5. Simplify the following. .

a) $14c + 6p + 3c = 17c + 6p$

b) $8f + 6f - 9j = 14f - 9j$

c) $-4b + 6h + 3b + 2h =$

$-8b + 8h$

d) $3q - 7q + 4j + 5j =$

$-4q + 9j$

6. Simplify:

(4)

a) $10a \times 4b = 40ab$

b) $2pq \times 6 \times r = 12pqr$

c) $3m \times -n \times 4 = -12mn$

d) $6a \times 3a^2 = 18a^3$

7. Simplify the following.

(4)

a) $4a \times bc \times 7d$
 $28abcd$

b) $3xy \times 8y$
 $24xy^2$

c) $\frac{25ef}{5e}$ $5f$

d) $\frac{20x^2y}{36xz}$ $\frac{5xy}{9z}$

8. Expand:

(3)

a) $3(4 + x)$ $12 + 3x$

b) $x(5y + 1)$

$5xy + x$

c) $3x(4y + 10)$

$12xy + 30x$

9. Simplify by expanding and collecting like terms:

(4)

a) $5(a + 2) + 3a$

$$5a + 10 + 3a$$
$$8a + 10$$

b) $a(3 + 4b) + 4(2a - ab)$

$$3a + 4ab + 8a - 4ab$$
$$11a.$$

10. Evaluate the following if "a" = 4 and "b" = 7

(4)

a) $a + b$

$$4 + 7 = 11$$

b) $10a + 7b$

$$10(4) + 7(7)$$
$$40 + 49 = 89$$

c) $-5a^2$

$$= -5(4)^2$$
$$= -80$$

d) $a^2 - b^2$

$$(4)^2 - (7)^2$$

11. Evaluate the following expressions when $d = -2$ and $e = 5$

(4)

a) $3de - 6$

$$3(-2)(5) - 6$$
$$-30 - 6 = -36$$

b) $5de + 100$

$$5(-2)(5) + 100$$
$$-50 + 100 = 50$$

c) $e^2 - 3d^2$

$$(5)^2 - 3(-2)^2$$
$$25 - 12 = 13$$

d) $2e^3 - 3d^2$

$$2(5)^3 - 3(-2)^2$$
$$250 - 12 = 238$$

12. Factorise fully.

(4)

a) $2x + 6$

$$2(x + 3)$$

b) $7a^2 + 14ab$

$$7a(a + b)$$

c) $24 - 16g$

$$8(3 - 2g)$$

d) $12x + 3xy$

$$3x(4 + y)$$

13. Two bus companies have different pricing structures.

(6)

Company A: \$120 call-out fee, plus \$80 per hour

Company B: \$80 call-out fee, plus \$100 per hour

- a) Write an expression for the total cost \$A of travelling n hours with company A.

$$\$A = 80n + \$120$$

- b) Write an expression for the total cost \$B of travelling for n hours with company

$$\$B = 100n + 80$$

- c) Hence state the cost of travelling for 3 hours with each company.

$$A = 80(3) + 120 = \$360$$

$$B = 100(3) + 80 = \$380$$

- d) For how long would you need to hire a bus to make company A the cheaper option?

$\$A$

n	0	1	1.5	2	3
$\$A$	120	200	240	280	360

n	0	1	1.5	2
$\$B$	80	180	230	280

Less than
2 hrs

- e) In the end, a school principal cannot decide which bus company to choose and hires 3 buses from company A and 2 buses from company B. Give an expanded expression to find the total cost for the school to hire the five buses for n hours.

$$3(80n + 120) + 2(100n + 80)$$

$$240n + 360 + 200n + 160$$

$$\$_{\text{Tot}} = 440n + 520$$

- f) If the trip lasts for 5 hours, how much does it cost to hire the five buses for this period of time?

$$\begin{aligned} \text{Tot} &= 440(5) + 520 \\ &= \$2720 \end{aligned}$$